# Chapter-1: Fundamentals of Vector Addition

**Vector:**

**Position Vector:**

**Commutative Law:**

**Associative Law:**

**Distribution Law:**

**Linear dependence and independence of a set vectors:**

**Q-4:** Show that the three vectors $\vec{A} = 2\hat{i} + \hat{j} - 3\hat{k}$, $\vec{B} = \hat{i} - 4\hat{k}$ and $\vec{C} = 4\hat{i} + 3\hat{j} - \hat{k}$ are linearly dependent. Determine a relation between them and hence show that the terminal points are collinear.

**Q-23:** Determine whether the vectors $\vec{r}_1 = 2\hat{i} - \hat{j} + \hat{k}$, $\vec{r}_2 = \hat{i} + 3\hat{j} - 2\hat{k}$, $\vec{r}_3 =- 2\hat{i} + \hat{j} - 3\hat{k}$ and $\vec{r}_4 = 3\hat{i} + 2\hat{j} + 5\hat{k}$ are linearly dependent or not.